Maths 4

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- Pascal's Triangle
- Output:

```
char ch = 'A';
int x = ch;
System.out.println(x);
```

- Output:

```
System.out.println('A' + 1);
```

- Output:
  1. ```
     int a = 5;
     double b = 2;
     System.out.println(a + b);
     ```
  2. `System.out.println(5 + 2 + "Hello");`

- Divide 10 by 3 and roundoff the answer upto 3 decimal Places.

1st → 1

2nd → 1 1

3rd → 1 2 1

4th → 1 3 3 1

5th → 1 4 6 4 1

0 1 2

0th row (1 element).

1st row (2 element).

2

3rd

$$^4C_2 = \frac{4!}{(4-2)!\times 2!} = \frac{4!}{2!\times 2!} = 6$$

$$\boxed{^nC_r = \frac{n!}{(n-r)!\,r!}}$$

$^3C_2 = 3$

$^nC_0 = 1$

col = 0

row = 0: 1

row 1: 1 1

row 2: 1 2 1

1 3 3 1

$^0C_0 = 1$

$^1C_0 = 1$, $^1C_1 = 1$

$^2C_0 = 1$ $^2C_1 = 2$

$^2C_2 = 1$

$^{10}C_3$

$$\to \frac{10!}{7!\times 3!} = \frac{10\times 9\times 8\times \cancel{7!}}{\cancel{7!}\times 3!}$$
     ```

$$= \frac{10 \times 9 \times 8}{1 \times 2 \times 3}$$

n = 10 $\quad {}^{10}C_3 \quad = \frac{10}{1} \times \frac{9}{2} \times \frac{8}{3} \quad = \frac{10 - i}{i + 1}$

$\boxed{\frac{10 - col}{col + 1}}$ n = 10, r = 3 $\rightarrow \frac{10 - 0}{0 + 1} \times \frac{10 - 1}{1 + 1} \times \frac{10 - 2}{2 + 1}$

3 1 3 3 1 $\boxed{n = 3}$

```
for (int i = 0; i < n; i++) {
    int ans = 1;
    for (int j = 0; j <= n; j++) {
        sout(ans);
        ans = ans * (i - j) / (j + 1);
    }
}
```

Type Conversion

byte int long float double

long b = 6L; double d = 3.3f

float a = 4.3f;

a, b, c, d, - - - - z.

```
for (char i = 'a'; i <= 'z'; i++) {
    sout(i);
}
```

for (int i = 0 ⟶ ~)